

# Shubh Pundir

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## Professional Summary

Applied AI Engineer specializing in production-grade LLM systems, real-time AI pipelines, and cost optimized GenAI architectures. Experienced in fine-tuning and developing GenAI services at scale with strong system design practices.

## Skills

**Core:** Java, Python, JavaScript, SQL

**AI/ML:** NLP, LLMs, Fine-Tuning(BERT, HuggingFace), RAG

**Systems:** FastAPI, Spring Boot, Microservices, Redis, Websockets, AWS, Docker, Prometheus, ELK, Unit Testing

## Experience

**Self** | *Independent Software Engineer*

Jan 2022 – Present

- Delivered 12+ production systems serving startups and SMEs across India, UK, Denmark, UAE, Morocco, and Ethiopia, building AI and backend systems that automate operations and scale with usage.
- Took end-to-end ownership of AI-enabled applications from architecture to deployment, monitoring, and post-launch iteration.
- Built a multi-tenant LLM support system handling 10K+ daily requests with Redis caching, request batching, and async workers to maintain low latency under peak load.

**MYTL Technologies** | *Founding Software Engineer Member*

June 2025 – Dec 2025

- Migrated legacy SIP system into a production-grade microservice ecosystem using Drachtio and FreeSWITCH, improving call reliability for 1,000+ active users.
- Reduced LLM inference cost by 40% by replacing full-context LLM calls with an intent-classification + RAG routing layer (F1: 0.93), cutting unnecessary token usage and improving latency
- Reducing real-time call latency by from 800ms to 300ms by optimizing event streaming and WebSocket handling for seamless OpenAI Realtime API integration.

**WDS: Wiz Digital Services** | *Machine Learning Engineer Intern*

Jan 2024 – Jun 2024

- Designed and deployed a privacy-preserving NLP API for PII detection and anonymization, achieving 92% accuracy and integrating it into Databricks-based, multi-client production pipelines.
- Developed conversational AI systems using Botpress and Rasa, automating issue resolution and ticket querying by ~78%.
- Annotated data records for NER Systems for 300+ documents with a custom parser for textual and image data.

## Projects

**Mudda - AI for civic bodies** | *Spring Boot, PostgreSQL, Flutter, LangGraph, Temporalio, QdrantDB*

Nov 2025

- Designed an Agentic AI-assisted workflow engine to replicate admin decision-making in auto-issue classification, de-duplications, AI voice calling staff, sending mails and maintaining an end-to-end audit history.
- Created a Redis Streams-based distributed execution layer to coordinate asynchronous jobs across independent microservices while ensuring reliability and scalability.
- Established architectural boundaries between AI cognition and operational execution by using LangGraph for policy-guided reasoning and Temporal for stateful workflow orchestration, enabling reliable, auditable, and deterministic automation of civic-service processes.

**Find-DB: Java based Database Engine from Scratch** | *Java, Maven*

April 2025

- Developed a lightweight custom binary file based database engine in Java with CRUD operations, indexing, data sharding to simulating core relational DBMS concepts.
- Accomplished logarithmic time lookup and prefix-based auto-completion by integrating a B+ Tree and Tries.

**AI-Based Music Recommendation System** | *Python, Airflow, Postgres, Spark, Pytorch, NLP, vLLM*

April 2025

- Built a medallion data pipeline combining audio feature extraction, NLP, AST analysis, etc. to generate rich representations of audio metadata for OLTP and OLAP purposes.
- Achieved 90% Precision@K by developing a hybrid recommendation logic in Pytorch using HuggingFace combining collaborative and content-based filtering with context awareness.
- Architected a multi-modal ensemble framework where independent NLP, CV, and numerical models were trained, evaluated, and fused through learned weighting to improve recommendation robustness and cold-start performance.

## Education

**The Northcap University**

2022 – 2026

*B.Tech in Computer Science Engineering*

GPA: 8.93/10.0

**Sri Sri Academy (12th ISC)**

2022

*PCM + Computer Science*

90.0%

**Sri Sri Academy (10th ICSE)**

2020

*Science + Computer Science*

94.7%

## Certifications

**Microsoft Azure AZ-900**

**Introduction to MongoDB**

**AWS Academy: Data Engineering**